EC-417 Introduction to Machine Intelligence

Term Project Report — Problem P25: Few-Shot Flowers (ViT vs. CNN)

Team Members: Pallavi (Roll No: 623106) & Geethika (Roll No: 623153)
Department: Electronics and Communication Engineering, NIT Andhra Pradesh
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1. Executive Summary & Introduction

Automatic classification of fine-grained botanical categories across a large number of classes remains a challenging task in computer vision due to high intra-class variance and subtle inter-class differences. In this term project for EC-417, we investigate Problem P25: Few-Shot Flowers using the Oxford Flowers-102 dataset (Nilsback & Zisserman, 2008). Under strict few-shot constraints with only 10 training images per class, we compare a conventional convolutional neural network baseline (ResNet-18) against a state-of-the-art Vision Transformer (ViT-B/16). Our evaluation demonstrates that pretrained self-attention architectures substantially outperform standard CNNs, achieving a test Top-1 accuracy of **95.84%** and Top-5 accuracy of **98.68%**.

2. Problem Framing & Dataset

The Oxford Flowers-102 dataset contains images of flower species belonging to 102 distinct categories. Each class contains a minimum of 40 samples, but the few-shot evaluation protocol provides only 1,020 training images (exactly 10 samples per class), 1,020 validation images, and 6,149 test images. This constrained data regime tests a model's ability to leverage transfer learning and inductive priors effectively.

Performance is evaluated using standard multi-class metrics:

- **Top-1 Accuracy:** The exact proportion of test images where the highest-probability prediction matches the true label.
- **Top-5 Accuracy:** The proportion where the true label is among the top 5 predicted categories, accommodating fine-grained ambiguities between visually similar floral species.

3. Methodology & Architecture

All input images were resized to 224×224 pixels and normalized using ImageNet channel statistics (mean: [0.485, 0.456, 0.406], std: [0.229, 0.224, 0.225]).

3.1 Baseline Model: ResNet-18

We utilized a standard ResNet-18 architecture pretrained on ImageNet (He et al., 2016). The final classification head was replaced with a fully connected linear layer mapping 512 feature dimensions to 102 output classes, trained using the Adam optimizer and Cross-Entropy loss.

3.2 Main Model: Vision Transformer (ViT-B/16)

As our primary model, we employed the pretrained Vision Transformer (`vit_base_patch16_224`) via the `timm` library (Dosovitskiy et al., 2020). ViT splits each image into non-overlapping 16×16 patches, projects them linearly into embedding space, adds learnable position embeddings, and processes the sequence through multiple stacked multi-head self-attention transformer blocks.

4. Experimental Results

Both models were trained on the official training split and evaluated on the held-out test set. Table 1 summarizes the quantitative performance achieved.

Model Architecture	Top-1 Accuracy (%)	Top-5 Accuracy (%)	Train Time
Baseline (ResNet-18)	78.50	91.74	~3 minutes
Main Model (ViT-B/16)	**95.84**	**98.68**	~5 minutes

5. Error Analysis & Discussion

The experimental results highlight the profound advantage of self-attention mechanisms in few-shot transfer learning. While ResNet-18 achieved a respectable 78.50% Top-1 accuracy, it struggled with fine-grained distinctions among closely related species where local texture alone is insufficient. In contrast, ViT-B/16 attained 95.84% Top-1 accuracy by capturing global contextual structures across the entire flower blossom.

6. Limitations & Ethics

Our models rely heavily on large-scale ImageNet pretraining priors. This study is intended strictly as an academic research exercise in machine intelligence and is not certified for automated botanical taxonomy or ecological deployment.

7. References

[1] Nilsback, M.-E., & Zisserman, A. (2008). Automated flower classification over a large number of classes. *ICVGIP*.

[2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *CVPR*.

[3] Dosovitskiy, A., et al. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*.